

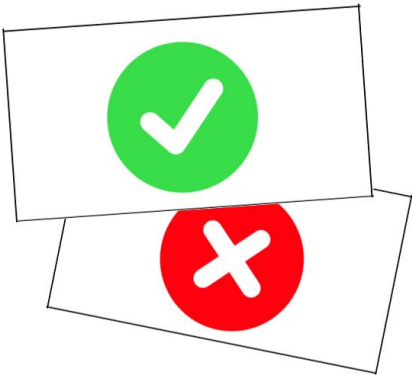
Simulation Activity Summary

_____ : A number that describes an aspect of a population

- N describes _____
- p describes _____
- μ describes _____

_____ : A number that is computed from sample data

- n is the _____
- $\hat{p}$ is the _____
- $\bar{x}$ is the _____



Sampling Variability:

- When taking a sample and gathering the sample proportion, we know that it will not be exactly equal to the population proportion due to sampling variability
- The idea that there is a _____ to samples

Population Distribution	Sample Distribution
• Only one population distribution • For proportions, this is categorical data that is broken up into “success” and “failure”	• Many sample distributions • Each sample distribution contains a $\hat{p}$ value, which represents the proportion of successes
N = Population Size = _____ p = proportion of success = _____	n = sample size = _____ $\hat{p}$ = proportion of successes = _____

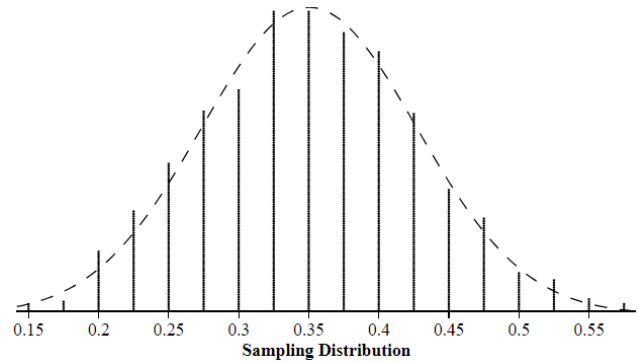
Sampling Distribution

- Only one sampling distribution
- The sampling distribution takes all possible $\hat{p}$ values from every possible sample distribution (of the same size n), and puts them together.
- This creates a unique sampling distribution with some common properties.

Mean of a Sampling Distribution for a Sample Proportion:

Standard Deviation of a Sampling Distribution for a Sample Proportion:

All possible samples of size $n = 40$ are found and the $\hat{p}$ values are put together in a distribution:



Mean = 0.35

Standard Deviation = 0.0754

Properties of a Sampling Distribution

Center of a Sampling Distribution

- When we start taking samples of a population, we have to revisit an earlier question in the year, how trustworthy is a statistic at being an estimate?
- A sampling procedure is _____ if it tends to systematically over-represent certain segments of the population and systematically under-represent others.
 - To avoid biased samples, we want to make sure that our sample uses _____.
- A statistic used to estimate a parameter is an _____ if the mean of its sampling distribution is equal to the value of the parameter being estimated:
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- In our example above:
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Shape of a Sampling Distribution

- No matter the shape of the population distribution (which itself is categorical and has no “shape”), the shape of the sampling distribution will always be _____ as long as a condition called the Large Counts Condition is satisfied.
- Large Counts Condition:
 -

- As long as n is large enough, we can say that the distribution will be approximately normal (never just “normal”). This is a consequence of what we call the Central Limit Theorem and we will discuss this more in the next section of notes!

Spread of a Sampling Distribution

- Statistics will vary from sample to sample due to sampling variability, however, larger samples have less variability than smaller samples.
- The spread of a sampling distribution DOES NOT depend on the _____, but instead, depends on the sample size.
- For independent samples (sampling with replacement) of a categorical variable from a population with a proportion p , the sample distribution has a standard deviation of
 -
- If you are sampling without replacement (like we did in our simulation activity), the standard deviation of the sampling distribution will be slightly different, HOWEVER, as long as a special rule is followed, this difference is negligible.
- This rule is called the _____, and we check it to make sure that sampling without replacement in a finite population will not affect the proportion of successes. As long as this condition is “satisfied”, we can use the above formula to calculate the standard deviation of our sampling distribution.
- 10% Condition:
 -
 - “Sample size is less than 10% of our population size”
 - If a problem does not mention a population size, we will want to describe N with words and will assume that it is large enough.
- In our example above:
 -
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- It is important to note that you only need to check the 10% condition when you are sampling from the population. When we have an experiment, we are not sampling, and therefore this condition does not need to be checked.

Example Problems for a Sampling Distribution for a Sample Proportion

A recent study looked at the percentage of young adult women who get less than 7 hours of sleep a night. In this study, they say that 45% of women get less than 7 hours of sleep a night. What is the probability that a random sample of 50 women will result in a sample proportion of 50% or higher?		
Step 1	Communicate the parameter in the context of the problem. Write down the probability you are trying to find.	
Step 2	Create your sampling distribution by finding the center, spread, and shape. Make sure to check each condition first.	
Step 3	Calculate the probability by converting your data to a z-score and using technology to find the probability. <i>*Drawing the curve is optional*</i>	
Step 4	Conclude the problem by answering the question in a complete sentence with the context of the problem.	

Example) Dane fails to study for his statistics final. The final has 100 multiple-choice questions, each with 5 choices. Assume all questions are independent and that Dane has given himself over to the Gods of Chance and randomly guesses on each of the 100 questions.

1) What is the parameter in the context of the problem?

2) What are the following values? $n =$ _____ and $p =$ _____

3) Is the sampling distribution approximately normal? Why or why not?

4) Is the sampling distribution unbiased? Why or why not?

5) Do we have to check the 10% condition here? Why or why not?

6) What are the following values?

$$\mu_{\hat{p}} =$$

$$\sigma_{\hat{p}} =$$

7) What is the probability that Dane will get between a 25% and 50% on the test? Show all work for this calculation.